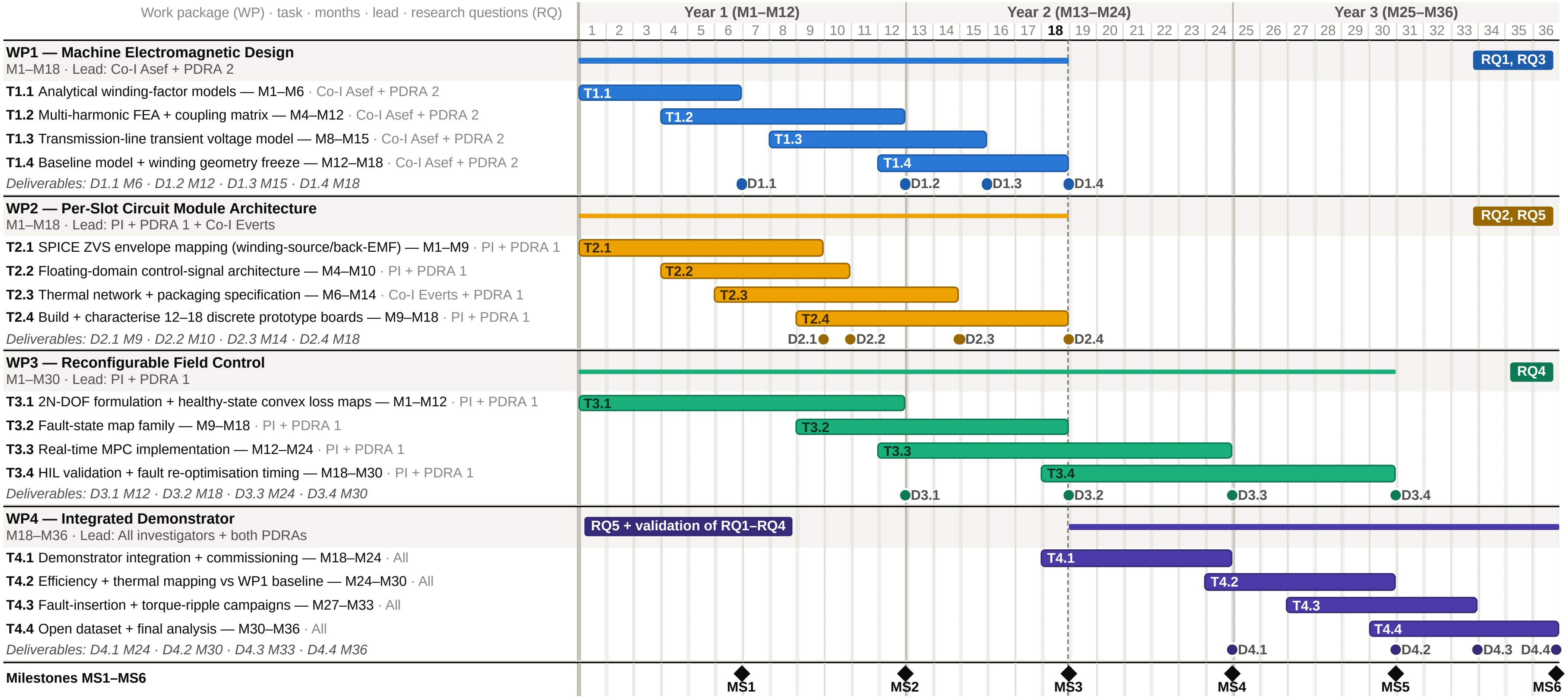
Foundations of Monolithically Integrated Per-Slot Machine Drives for Reconfigurable Aerospace Propulsion — Project Plan (36 months)

Principal Investigator (PI): Dr Mehdi Baghdadi, UCL Advanced Propulsion Laboratory · Co-Investigators (Co-Is): Dr Pedram Asef, Dr Marilize Everts · two Postdoctoral Research Associates (PDRA 1, PDRA 2) · Technology Readiness Level (TRL) 1–3

[PI TO CONFIRM: task/deliverable/milestone skeleton proposed by drafting team]



WP1 Machine electromagnetic design (RQ1, RQ3) · WP2 Per-slot circuit module architecture (RQ2, RQ5) · WP3 Reconfigurable field control (RQ4) · WP4 Integrated demonstrator (RQ5 + validation of RQ1–RQ4)

● Deliverable Dn.k · ◆ Milestone · ---- M18 gate: MS3 design freeze and integration go/no-go

Milestones: MS1 (M6) WP1–WP2 interface parameter freeze · **MS2** (M12) validated models exchanged across WPs · **MS3** (M18) design freeze and integration go/no-go · **MS4** (M24) demonstrator commissioned · **MS5** (M30) control validated on hardware · **MS6** (M36) dataset released, programme complete.

Dependencies: WP1 and WP2 run in parallel from month 1, exchanging parameterised interface models at MS1 and validated models at MS2. WP3 begins analytical formulation from month 1 using parameterised placeholder models, and consumes validated WP1 and WP2 outputs as they arrive. Only WP4 is dependent: it starts at M18, gated by the MS3 design freeze and integration go/no-go. The programme contains no sequential chain of work packages — three of the four strands are live from the first month.

Abbreviations: WP work package · RQ research question · M project month · Dn.k deliverable · MSn milestone · PI Principal Investigator · Co-I Co-Investigator · PDRA Postdoctoral Research Associate · FEA finite element analysis · SPICE Simulation Program with Integrated Circuit Emphasis · ZVS zero-voltage switching · back-EMF back electromotive force · 2N-DOF 2N degrees of freedom for N driven sections · MPC model predictive control · HIL hardware-in-the-loop · TRL Technology Readiness Level.